EXPERIMENT 6: PROGRAMS IN PROLOG

AIM

To implement family relationships, animal classification, and simple arithmetic using Prolog facts, rules, and queries.

PROCEDURE

1. Define facts and rules in Prolog for each problem.

2. Enter the required query for each knowledge base.

3. Execute the query and observe the result returned by Prolog.

CODE

1. Family Relationships

```
% Facts
parent(john, mary).
parent(mary, susan).
parent(john, mike).

% Rule
grandparent(X, Y) :- parent(X, Z), parent(Z, Y).

% Query
?- grandparent(john, susan).
```

Output:

```
true
```

2. Animal Classification

```
% Facts
has_hair(tiger).
gives_milk(tiger).

% Rule
mammal(X) :- has_hair(X), gives_milk(X).

% Query
?- mammal(tiger).
```

Output:

```
true.
```

3. Simple Arithmetic Knowledge Base

```
% Rule
add(X, Y, Z) :- Z is X + Y.

% Query
?- add(5, 7, Z).
```

Output:

```
Z = 12.
```

RESULT

Thus, the Prolog programs for family relationships, animal classification, and simple arithmetic were executed successfully.